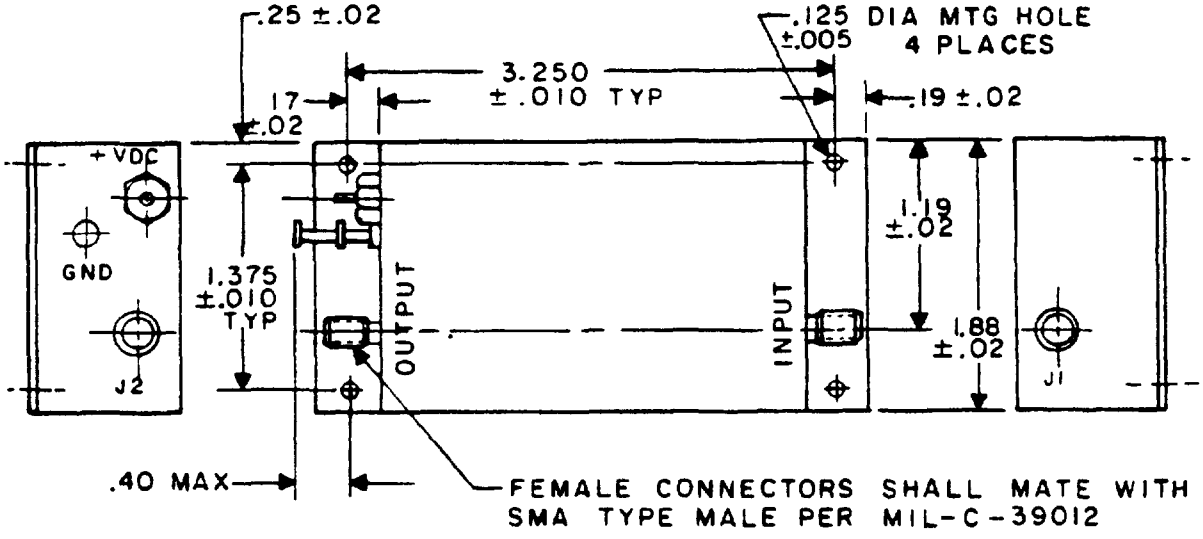
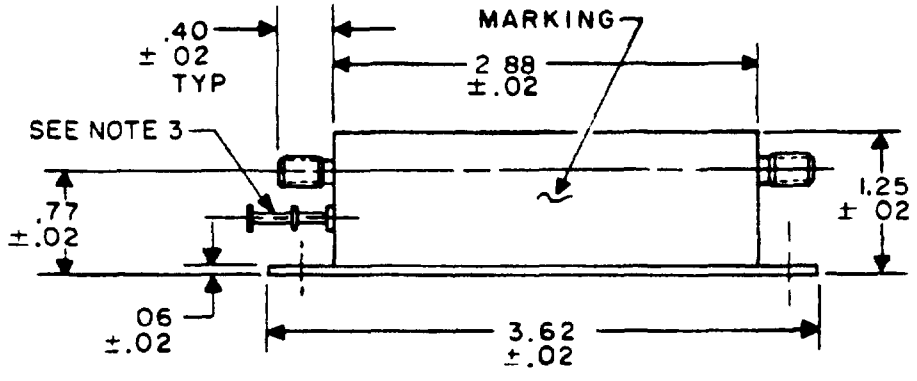
MILITARY SPECIFICATION SHEET

AMPLIFIERS, RADIOFREQUENCY AND MICROWAVE, SOLID-STATE, NARROWBAND, COAXIAL CONFIGURATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring amplifiers described herein shall consist of this specification sheet and the latest issue of MIL-A-28875.





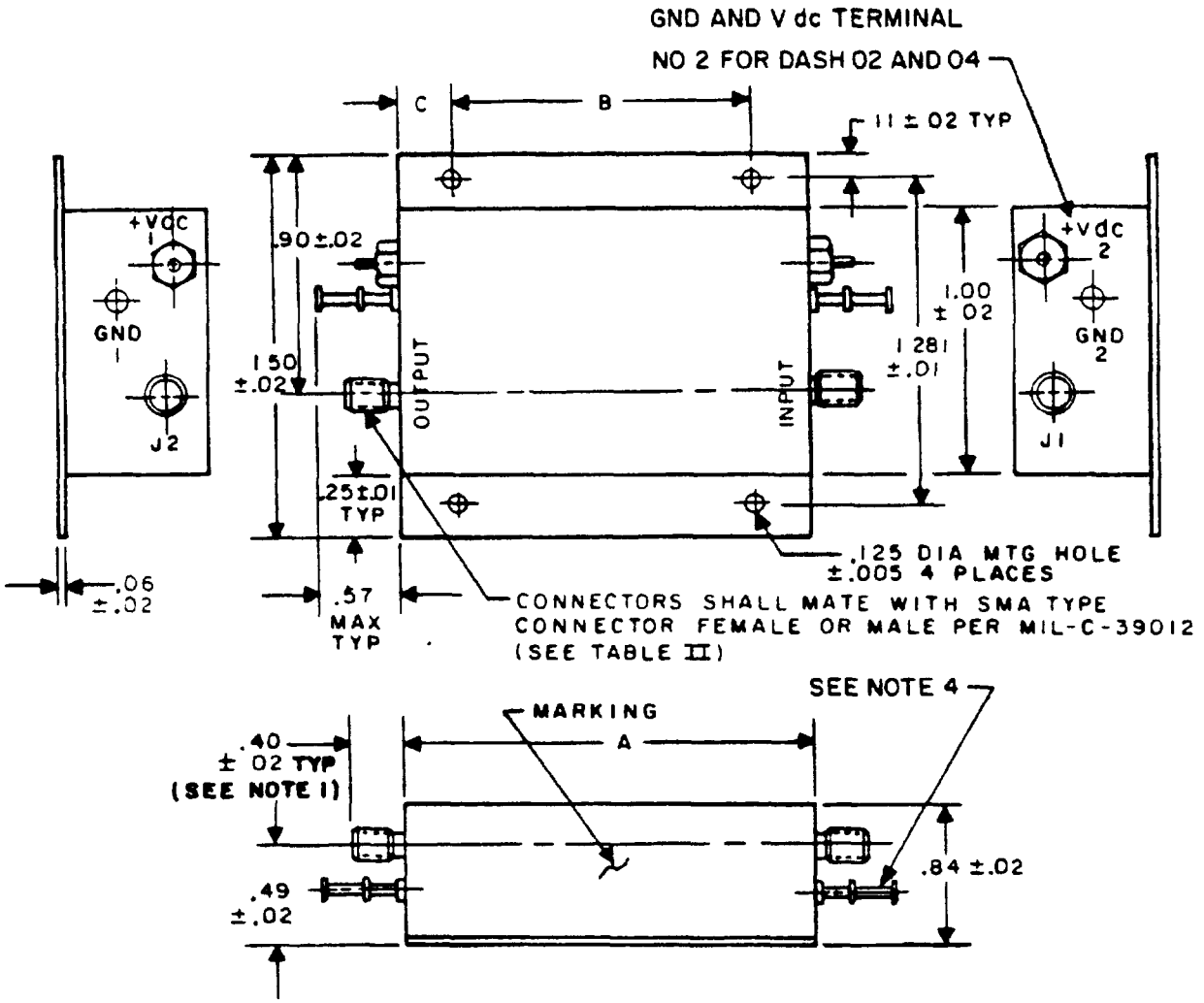
INCHES	MM
.005	.13
.010	.25
.02	.5
.06	1.5
.125	3.18
.17	4.3
.19	4.8
.25	6.4
.40	10.2
.77	19.6
1.19	30.2
1.25	31.8
1.375	34.92
1.88	47.8
2.88	73.2
3.250	82.55
3.62	92.0

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Barrel diameter shall accept two turns of AWG NO. 24 wire.

FIGURE 1. Dimensions and configuration, dash number -02.

FSC 5985

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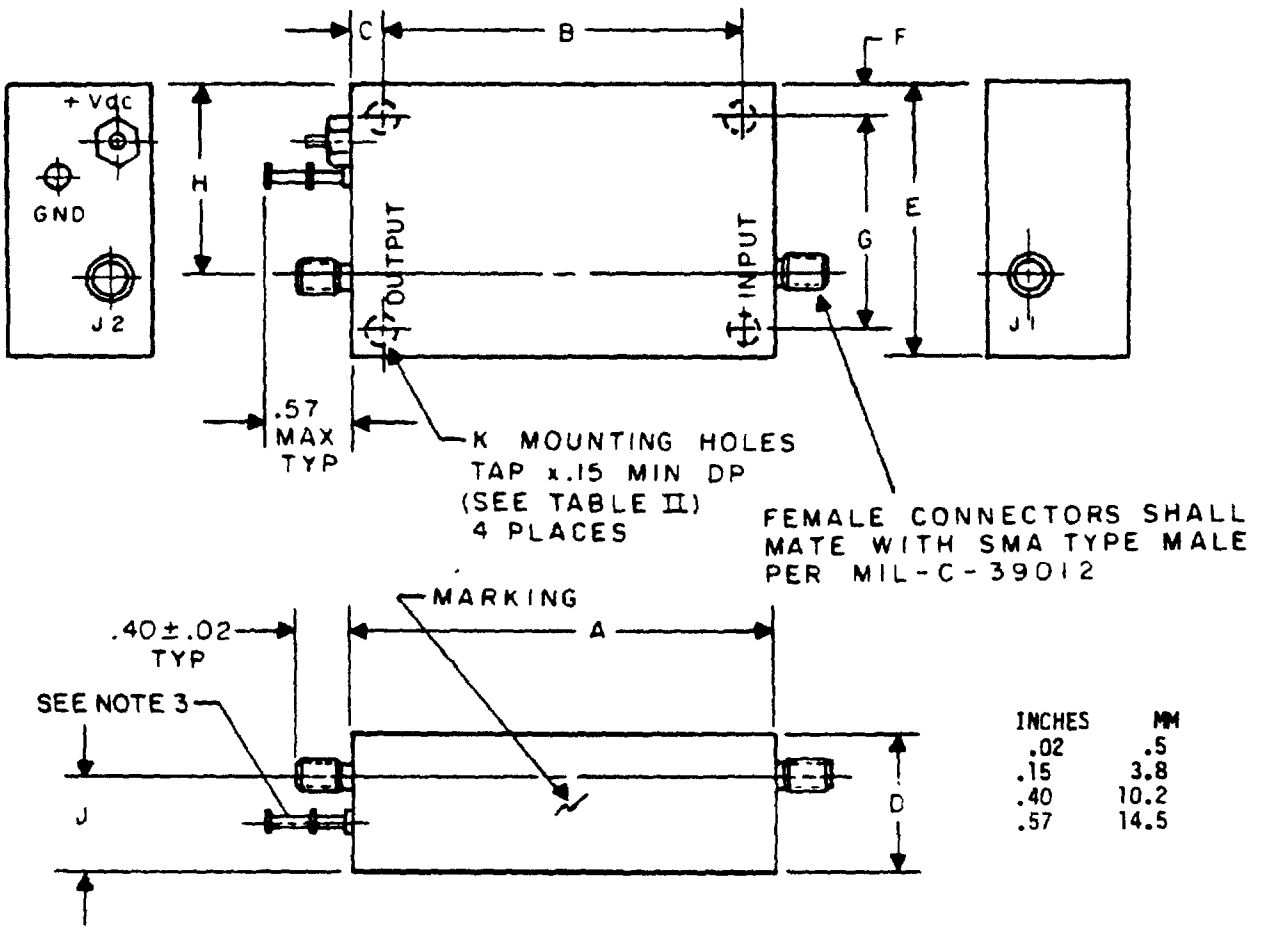


INCHES	MM	INCHES	MM
.005	.13	.49	12.5
.01	.3	.57	14.5
.02	.5	.84	21.3
.06	1.5	.90	22.3
.11	2.8	1.00	25.4
.125	3.18	1.281	32.54
.19	4.8	1.37	34.8
.20	5.1	1.50	38.1
.25	6.4	2.00	50.8
.40	10.2	2.40	61.0

NOTES:
1. Female connector shown, for male connectors allow .75 max.
2. Dimensions are in inches.
3. Metric equivalents are given for general information only.
4. Barrel diameter shall accept two turns of AWG NO. 24 wire.

FIGURE 2. Dimensions and configuration, dash numbers -03, -04, -05, -08, and -09.

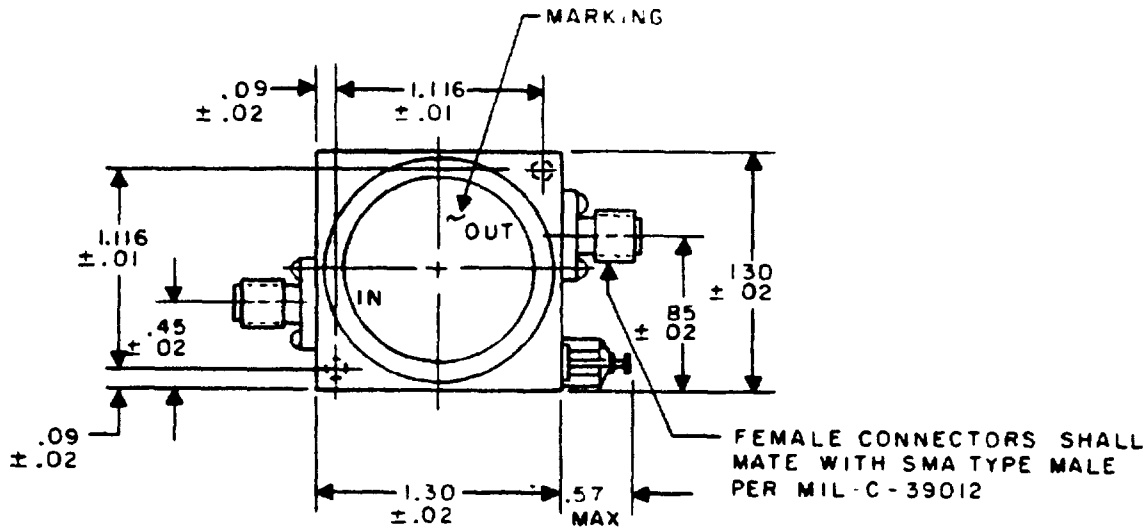
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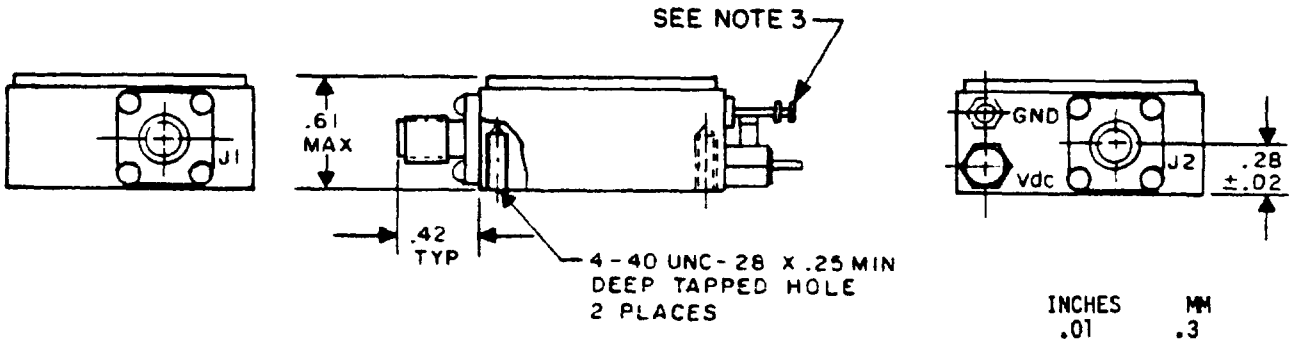


NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Barrel diameter shall accept two turns of AWG NO. 24 wire.

FIGURE 3. Dimensions and configuration, dash numbers -01, -06, -07, and -13.

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INCHES	MM
.01	.3
.02	.5
.09	2.3
.25	6.4
.28	7.1
.42	10.7
.45	11.4
.57	14.5
.61	15.5
.85	21.6
1.116	28.35
1.30	33.0

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Barrel diameter shall accept two turns of AWG NO. 24 wire.

FIGURE 4. Dimensions and configuration, dash numbers -10, -11, and -12.

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TABLE 1. Electrical characteristics (25°C unless otherwise specified).

Dash number	Frequency range	DC input power				Gain (dB) Min	Gain flatness (dB) Max	RF output power (dBm) Min	VSWR Max		Noise figure (dB) Max	Intercept point 3rd order (dBm) Min	Harmonic rejection at output (dBc)	Gain vs temperature (dB)
		Input No. 1		Input No. 2					IN	OUT				
		Voltage	Current Max (mA)	Voltage	Current Max									
01 S1 S2 N	32.5-42.5 MHz	28 ±1%	250	N/A	N/A	21.0	±0.5	25	2.0.1	2.0·1	12	+35	-70	---
02 S1 S2 N	220-320 MHz	15 ±1%	50	N/A	N/A	16.5	±0.5	-13	1.5:1	2.0 1	4	---	---	±1
03 S1 S2 N	300-565 MHz	15 ±1%	130	N/A	N/A	24 23 at +72°C	±1.0	+14 +13° at +72°C	2.0:1	2.0:1	6.5 7 at +72°C	N/A	N/A	N/A
04 S1 S2 N	300-565 MHz	15 ±1%	30	N/A	N/A	14	±1.0	+ 7	2.0:1	2.0 1	6.5	N/A	N/A	N/A
05 S1 S2 N	400-500 MHz	24 ±1%	225	N/A	N/A	35	N/A	N/A	2.0:1	2.0 1	6 6.5 at +72°C	33 31 at +72°C	N/A	+2.5 at -55° -2.5 at 72°
06 S1 S2 N	475-610 MHz	12 ±5%	30	N/A	N/A	15	±0.5	+10	2.0:1	2.0 1	6	N/A	N/A	N/A
07 S1 S2 N	475-610 MHz	15 ±5%	30	N/A	N/A	15	±0.5	+10	2.0.1	2.0 1	6	N/A	N/A	N/A
08 S1 S2 N	950-1250 MHz	24 ±1%	65	15 ±1%	55 mA	34	±0.5	+11	2.5:1	2.5·1	5	N/A	N/A	N/A

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TABLE I. Electrical characteristics (25°C unless otherwise specified) - Continued.

Dash number	Frequency range	DC input power				Gain (dB) Min	Gain Flatness (dB) Max	RF output power (dBm) Min	VSWR Max		Noise figure (dB) Max	Intercept point 3rd order (dBm) Min	Harmonic rejection at output (dBc)	Gain stability (dB)
		Input No. 1		Input No. 2					IN	OUT				
		Voltage	Current Max	Voltage	Current Max									
S1 09 S2 N	1.35-1.62 GHz	24 ±1%	65 mA	15 ±1%	40 mA	14 13 at +72°C	±1.5	+11 +10° at +72°C	2.5 1	2.5:1	8.5 9 at +72°C	N/A	N/A	N/A
S1 10 S2 N	2.9-3.1 GHz	15 ±1%	100 mA	N/A	N/A	10	±0.5	+6.5	1.5 1	1.75:1	20	N/A	N/A	N/A
S1 11 S2 N	3 - 4 GHz	15 ±1%	120 mA	N/A	N/A	11 from 0°C-+85°C 10 from -54°C-+85°C	+0.5 from 0°C-+85°C ±1.0 from -54°C-+85°C	+10 from 0°C-+85°C 8 from -54°C-+85°C	1 5·1	1.5:1	9.5 from +65°C-+85°C	N/A	N/A	N/A
S1 12 S2 N	3 - 4 GHz	15 ±1%	400 mA	N/A	N/A	10 from -54°C-+85°C	+0.5 from 0°C-+85°C ±1.0 from -54°C-+85°C	+20 from 0°C-+85°C +18 from -54°C-+85°C	1.5:1	1.5:1	12 from +65°C-+85°C	N/A	N/A	N/A
S1 13 S2 N	11.7-12.2 GHz	12 ±0.5%	180 mA	N/A	N/A	30	±0.5	+10	2.0:1	2.0·1	12	N/A	N/A	N/A

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TABLE II. Physical and environmental characteristics.

Dash number	Dimensions										Weight (LB) Max	Temperature range		Seal	Salt	EMI	RF connectors		Figure number
	A	B	C	D	E	F	G	H	J	K		Operating case temp	Nonoperating				Input	Output	
01 N S1 S2	3.00 ±.02	2.500 ±.01	.25 ±.02	.80 ±.02	1.75 ±.02	.06 ±.02	1.625 ±.01	.88 ±.02	.50 ±.02	.086-56 UNC-2B	0.33	0°C-+55°C	-55°C +100°C	Non-hermetic	N/A	N/A	Female	Female	3
02 N S1 S2	---	---	---	---	---	---	---	---	---	---	1.1	0°C-+65°C	-62°C-+72°C	Non-hermetic	Applicable	CE03 of MIL-STD-461	Female	Female	1
03 S1 S2 N	1.37 ± 02	1 000 ± 01	.19 ±.02	---	---	---	---	---	---	---	0 18	-55°C-+72°C	-57°C-+81°C	Non-hermetic	Applicable	CE03 of MIL-STD-461	Male	Female	2
04 S1 S2 N	1.37 ±.02	1.000 ± 01	.19 ±.02	---	---	---	---	---	---	---	0.18	-55°C-+72°C	-57°C-+81°C	Non-hermetic	Applicable	CE03 of MIL-STD-461	Male	Female	2
05 S1 S2 N	2.40 ±.02	2.000 ±.01	.20 ±.02	---	---	---	---	---	---	---	0.30	-55°C-+72°C	-57°C-+81°C	Non-hermetic	Applicable	CE03 of MIL-STD-461	Female	Female	2
06 S1 S2 N	1.25 ±.02	1.000 ± 01	.10 ±.02	.80 ±.02	1.25 ±.02	.225 ±.01	.800 ±.01	.625 ±.01	.295 ±.01	2-56 UNC-2B	0.12	-55°C-+95°C	-62°C-+95°C	Non-hermetic	Applicable	CE03 of MIL-STD-461	Female	Female	3
07 S1 S2 N	1.25 ±.02	1 000 ±.01	.10 ±.02	.80 ±.02	1.25 ±.02	.225 ±.01	.800 ±.01	.625 ±.01	.295 ±.01	2-56 UNC-2B	0.12	-55°C-+95°C	-62°C-+95°C	Non-hermetic	Applicable	CE03 of MIL-STD-461	Female	Female	3
08 S1 S2 N	2.40 ±.01	2.000 ±.01	.20 ±.02	---	---	---	---	---	---	---	0.30	-55°C-+72°C	-62°C-+81°C	Non-hermetic	Applicable	CE03 of MIL-STD-461	Female	Male	2
09 S1 S2 N	1.37	1.000	.19 ±.02	---	---	---	---	---	---	---	0.18	-55°C-+72°C	-62°C-+81°C	Non-hermetic	N/A	CE03 of MIL-STD-461	Female	Female	2

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TABLE II. Physical and environmental characteristics - Continued.

Dash number	Dimensions										Weight (LB) Max	Temperature range		Seal	Salt	EMI	RF Connectors		Figure number
	A	B	C	D	E	F	G	H	J	K		Operating case temp	Nonoperating				Input	Output	
10 S1	---	---	---	---	---	---	---	---	---	---	0.24	0°C- +65°C	-62°C- +72°C	Non-hermetic	Applicable	CE03 of MIL-STD-461	Female	Female	4
10 S2												-54°C +85°C	-54°C- +100°C						
10 N																			
11 S1																			
11 S2																			
11 N																			
12 S1																			
12 S2																			
12 N																			
13 S1	2.52 ±.02	2.275 ±.01	.12 ±.02	.88 ±.02	1.00 ±.02	.12 ±.02	.760 ±.01	.35 ±.02	.43 ±.02	4-40 UNC-2B	0.24	-54°C- +85°C	-54°C +85°C	Non-hermetic	N/A	CE03 of MIL-STD-461	Female	Female	3
13 S2																			
13 N																			

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REQUIREMENTS.

Dimensions and configuration: See figures 1, 2, 3, 4, and table II.

Electrical characteristics: See table I.

Physical and environmental characteristics: See table II.

Part number: M28875/2-(dash number from tables I and II).

NOTE: Amplifiers covered by this specification sheet are substitutable for the following manufacturer's or user's part number. This information in no way implies that the manufacturer's part is suitable as a substitute for the military part number.

Dash number	Manufacturer's code	Manufacturer's part number
01S	18876	13052368
01S	51025	HD4A3501
02S	05869	723854-2
02S	24539	SD6-0475M
03S	05869	723129-5
03S	24539	SD7-1187M
04S	05869	723129-2
04S	24539	SD7-1184M
05S	05869	723129-6
05S	24539	SD7-1054M
06S	19156	920033-001
06S	51025	4923PS-1
07S	19156	920039-001
07S	51025	5923PS-1
08S	05869	723129-1
08S	24539	SD7-1183M
09S	05869	723129-3
09S	24539	SD7-1185M
10S	05869	723855-1
10S	24539	ASF6811
11S	97942	582R593-02
11S	24539	ASF8930
12S	97942	582R593-03
12S	24539	ASF8929
13S	24587	0413031-1
13S	51025	XM183301

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AV
Navy - AS, MC
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5985-0968-2)

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